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References

Browne: Precision of prediction **Browne-1969**

Michael W. Browne. "Precision of prediction". In: *ETS Research Bulletin Series* 1969.2 (Dec. 1969). ISSN: 0424-6144. DOI: 10.1002/j.2333-8504.1969.tb00748.x.

Abstract: Precision of prediction in multiple linear regression is examined. Two measures of predictive precision, predictive mean square error, d^2, and the squared weight validity, w^2, are employed. The use of an existing estimator of $\epsilon(d^2)$ as an estimator of d^2 is proposed and the mean squared error of estimation of this estimator about d^2 is obtained. A significance test is given. Estimators of w^2 are derived and an asymptotic approximation for their variances is given. These estimators of w^2 are functions of estimators of ρ^2, the squared multiple correlation coefficient, and of ρ^4. The bias and mean squared error of estimation of some known estimators of ρ^2 and of a proposed estimator of ρ^4 are examined. Monte Carlo experiments are used to compare the proposed estimators of w^2 with an estimator due to Burket [1964]. An efficient procedure for generating w^2 and the estimates of w^2 is described. The mean squared errors of estimation of cross-validation estimators of d^2 and w^2 are obtained and disadvantages of the cross-validation procedure are discussed. An example is used to illustrate relationships between predictive precision and the number of predictors. The paper is primarily concerned with a random predictor model but results for a fixed predictor model are also given.

Duncan: Some linear models for two-wave, two-variable panel analysis **Duncan-1969**

Otis D. Duncan. "Some linear models for two-wave, two-variable panel analysis". In: *Psychological Bulletin* 72.3 (Sept. 1969), pp. 177–182. ISSN: 0033-2909. DOI: 10.1037/h0027876.

Abstract: In the absence of a sufficient number of a priori substantive assumptions ruling out certain conceivable causal linkages among variables, neither cross-lagged correlation nor any other technique for analyzing 2-wave, 2-variable panel data will yield a unique causal inference. A wide variety of distinct linear causal models will always be compatible with a given set of panel data.

Goodman: On the exact variance of products **Goodman-1960**

Leo A. Goodman. "On the exact variance of products". In: *Journal of the American Statistical Association* 55.292 (Dec. 1960), pp. 708–713. DOI: 10.1080/01621459.1960.10483369.

Abstract: A simple exact formula for the variance of the product of two random variables, say, x and y, is given as a function of the means and central product-moments of x and y. The usual approximate variance formula for xy is compared with this exact formula; e.g., we note, in the special case where x and y are independent, that the "variance" computed by the approximate formula is less than the exact variance, and that the accuracy of the approximation depends on the sum of the reciprocals of the squared coefficients of variation of x and y. The case where x and y need not be independent is also studied, and exact variance formulas are presented for several different "product estimates." (The usefulness of exact formulas becomes apparent when the variances of these estimates are compared.) When x and y are independent, simple unbiased estimates of these exact variances are suggested; in the more general case, consistent estimates are presented.

Granger: Investigating causal relations by econometric models and cross-spectral methods **Granger-1969**

C. W. J. Granger. "Investigating causal relations by econometric models and cross-spectral methods". In: *Econometrica* 37.3 (Aug. 1969), p. 424. ISSN: 0012-9682. DOI: 10.2307/1912791.

Abstract: There occurs on some occasions a difficulty in deciding the direction of causality between two related variables and also whether or not feedback is occurring. Testable definitions of causality and feedback are proposed and illustrated by use of simple two-variable models. The important

problem of apparent instantaneous causality is discussed and it is suggested that the problem often arises due to slowness in recording information or because a sufficiently wide class of possible causal variables has not been used. It can be shown that the cross spectrum between two variables can be decomposed into two parts, each relating to a single causal arm of a feedback situation. Measures of causal lag and causal strength can then be constructed. A generalisation of this result with the partial cross spectrum is suggested.

Hamilton: A rating scale for depression **Hamilton-1960**

Max Hamilton. "A rating scale for depression". In: *Journal of Neurology, Neurosurgery & Psychiatry* 23.1 (Feb. 1960), pp. 56–62. ISSN: 0022-3050. DOI: 10.1136/jnnp.23.1.56.

Kalman: A new approach to linear filtering and prediction problems **Kalman-1960**

R. E. Kalman. "A new approach to linear filtering and prediction problems". In: *Journal of Basic Engineering* 82.1 (Mar. 1960), pp. 35–45. DOI: 10.1115/1.3662552.

Abstract: The classical filtering and prediction problem is re-examined using the Bode-Shannon representation of random processes and the "state-transition" method of analysis of dynamic systems. New results are: (1) The formulation and methods of solution of the problem apply without modification to stationary and nonstationary statistics and to growing-memory and infinite-memory filters. (2) A nonlinear difference (or differential) equation is derived for the covariance matrix of the optimal estimation error. From the solution of this equation the co-efficients of the difference (or differential) equation of the optimal linear filter are obtained without further calculations. (3) The filtering problem is shown to be the dual of the noise-free regulator problem. The new method developed here is applied to two well-known problems, confirming and extending earlier results. The discussion is largely self-contained and proceeds from first principles; basic concepts of the theory of random processes are reviewed in the Appendix.